

Reservoir Computing Model for Liquid Crystal Cell Responsive to Design Parameter and Its Augmentation With Novel Transfer Learning Framework

Makoto Watanabe^{ID}, Graduate Student Member, IEEE, Reo Otsuki^{ID}, Kiyoshi Kotani, and Yasuhiko Jimbo^{ID}, Member, IEEE

Abstract—Macro models of liquid crystal cells that prioritize input–output responses over physical accuracy are high-speed and valuable for large-scale circuit simulations. However, the simplicity of the macro model makes it challenging to achieve accurate responses when design parameters, such as the electrode width, change. In this study, we have developed a macro model of liquid crystal cells that follows changes in design parameters using “reservoir computing (RC),” one of the machine-learning (ML) methods. We have obtained high accuracy with trained and untrained design values by adding a port to provide a design value and by adjusting the hyperparameters that significantly affect prediction accuracy. In addition, we demonstrated that connecting a well-trained model with a specific design value in series or parallel to the reservoir has achieved equivalent prediction ability even when trained with a smaller amount of data. This new transfer learning (TL) approach in RC can be used even when the dimension of the output weight matrix changes, which has not been covered in previous studies. Display designers can use our model in circuit simulators to optimize design values while estimating the impact of design values on optical properties.

Index Terms—Liquid crystal, machine learning (ML), macro model, reservoir computing (RC), transfer learning (TL).

I. INTRODUCTION

THE performance of liquid crystal displays (LCDs) is expected to improve further as markets for augmented reality, virtual reality, and in-vehicle applications expand. Therefore, new materials and cell structures are being actively developed [1], [2], [3], and there are many cases in which

circuit simulations are performed with different design values depending on the product specifications. To obtain highly accurate results in circuit simulations, liquid crystal cells that exhibit nonlinear and time-dependent behaviors can be represented as macro models and incorporated into these simulations. However, in the macro models reported thus far [4], [5], [6], a liquid crystal is considered as a single conceptual threaded spring, and the forces acting on this liquid crystal, such as the electric field, elasticity, and viscosity, can be roughly approximated using simple equations with minimum physical constants. These macro models must be updated when liquid crystal cells include new phenomena. Developing macro models based on conventional methods is typically time-consuming, and inventions are occasionally required.

The rapid progress of machine-learning (ML) methods in recent years has allowed predictions from big data. Among various ML methods, reservoir computing (RC) [7] can handle dynamic systems and is suitable for circuit simulations because it has a simple framework and works very fast. RC also requires less computational cost for training than other recurrent neural networks because the weights of the input and reservoir are fixed, and only the output weights are learned. RC has been employed to predict spatiotemporally chaotic systems [8] and perform fluid analysis [9]. In our previous study [10], we proposed using RC to develop a macro model of liquid crystal cells containing one of the complex physical phenomena, the flexo electric effect [11], and to use it in circuit simulations. However, our RC model did not follow changes in the design values.

Preparing big data in advance is another critical issue when developing macro models via ML. In particular, a considerable amount of data must be prepared with multiple design values to train the RC model to follow the changes in design values. In this context, transfer learning (TL) can ensure high prediction accuracy with a small amount of data by utilizing a high-accuracy pretrained model. Previous studies [12], [13], [14] have incorporated TL with RC, which arithmetically adds a correction weight matrix to the already trained output weight matrix. Therefore, it is necessary to make the dimensions of these weight matrices identical.

Received 27 August 2024; revised 12 October 2024; accepted 2 December 2024. Date of publication 13 December 2024; date of current version 24 January 2025. This work was supported in part by KAKENHI under Grant 22K19785, in part by the Asahi Glass Foundation, and in part by The Precise Measurement Technology Promotion Foundation. The review of this article was arranged by Editor X. Guo. (Corresponding author: Makoto Watanabe.)

Makoto Watanabe is with the Graduate School of Engineering, The University of Tokyo, Tokyo 153-8904, Japan (e-mail: watanabe@neuron.t.u-tokyo.ac.jp).

Reo Otsuki and Kiyoshi Kotani are with the Graduate School of Frontier Sciences, The University of Tokyo, Chiba 277-8563, Japan.

Yasuhiko Jimbo is with the Graduate School of Engineering, The University of Tokyo, Tokyo 113-8656, Japan.

Digital Object Identifier 10.1109/TED.2024.3512479

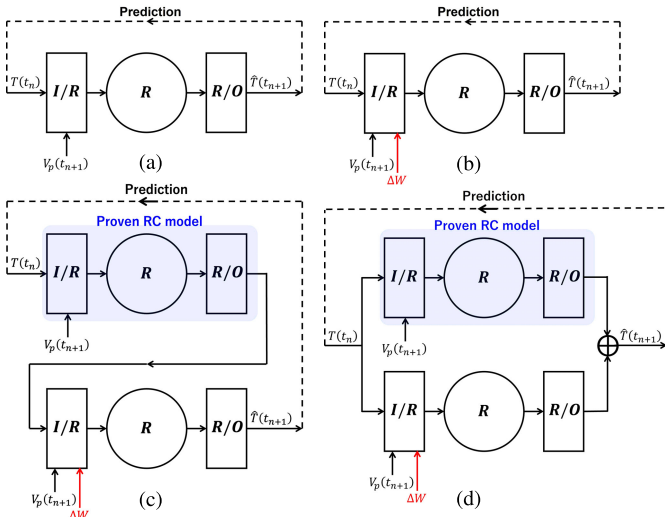


Fig. 1. RC model configuration. (a) Conventional RC model. (b) Single RC model. (c) Series-connected multiple RC model. (d) Parallel-connected multiple RC model.

However, when developing an RC model that can respond to various design values, the dimension of the weight matrix increases along with the number of input ports in the RC model, making previous methods inapplicable.

The first purpose of this study is to develop a macro model of liquid crystal cells that can respond to design values using the RC. We equipped a conventional RC model with a port to input a design value and adjusted the hyperparameters, which significantly affected the prediction accuracy. The second purpose of this study is to explore a TL approach that can be applied even when the dimension of the output weight matrix in RC is changed. We hypothesized that TL could be achieved by connecting an RC model to be optimized to a proven RC model that was already well-trained for a specific design value.

II. METHODOLOGY

A. RC Model

Fig. 1(a) shows a conventional RC model of liquid crystal cells without inputs of design parameters. I/R and R/O denote the input-to-reservoir and reservoir-to-output couplers, respectively. R denotes the reservoir composed of D_r -dimensional vector $\mathbf{r}(t_n) = [r_1(t_n), r_2(t_n), \dots, r_{D_r}(t_n)]^T$ and initialized as $\mathbf{r}(t_1) = \mathbf{0}$. We derive explicit formulas using the echo state network (ESN) [15].

In the training phase, I/R receives the transmittance value for the current time $T(t_n)$ and a pixel electrode voltage for the next time $V_p(t_{n+1})$. These input variables to the RC model are grouped and represented as the input state vector $\mathbf{u}_{in}(t_n) = \begin{bmatrix} V_p(t_{n+1}) \\ T(t_n) \end{bmatrix}$.

Next, the value of the state in the reservoir $\mathbf{r}(t_n)$ is updated with

$$\mathbf{r}(t_{n+1}) = (1 - a)\mathbf{r}(t_n) + a \tanh\left(\mathbf{W}_{res}\mathbf{r}(t_n) + \mathbf{W}_{in}\begin{bmatrix} 1 \\ \mathbf{u}_{in}(t_n) \end{bmatrix}\right) \quad (1)$$

where a represents the leaking rate, which can be considered the speed of the reservoir update. The elements of $\mathbf{W}_{res}$ with a dimension of $D_r \times D_r$ were generated randomly from a uniform distribution in $[-\sigma, +\sigma]$, where $\sigma > 0$ was adjusted so that the maximum absolute value of the eigenvalue of $\mathbf{W}_{res}$ was ρ . The quantity ρ represents the spectral radius of the reservoir, which is regarded as the memory length of the reservoir. The quantity d_i ($0 < d_i \leq 1$) represents the sparse density, which sets the number of nonzero elements in each row of $\mathbf{W}_{res}$ to the value obtained by converting the product of D_r and d_i to an integer value by truncation. The elements of the input weight matrix $\mathbf{W}_{in}$ with a dimension of $D_r \times 3$ that functioned as I/R were selected randomly from a uniform distribution in $[-0.5, +0.5]$, which is the same range adopted in our previous study on an RC model of liquid crystal cells [10]. Finally, the predicted value for the next time step $\hat{T}(t_{n+1})$ is calculated as follows:

$$\hat{T}(t_{n+1}) = \mathbf{W}_{out} \begin{bmatrix} 1 \\ \mathbf{u}_{in}(t_n) \\ \mathbf{r}(t_{n+1}) \end{bmatrix} \quad (2)$$

where $\mathbf{W}_{out}$ with a dimension of $1 \times (D_r + 3)$ represents the output weight matrix, which functions as R/O . The objective of the training phase was to find the components of $\mathbf{W}_{out}$ so that the output $\hat{T}(t_{n+1})$ from R/O was asymptotic to the target $T(t_{n+1})$ in the training data. The components of $\mathbf{W}_{out}$ were computed as follows [16]:

$$\mathbf{W}_{out} = \mathbf{Y}\mathbf{R}^T(\mathbf{R}\mathbf{R}^T + \lambda\mathbf{I})^{-1} \quad (3)$$

where λ denotes the regularization coefficient in Ridge regression to avoid overfitting, and $\mathbf{I}$ represents an identity matrix. In (3), $\mathbf{Y}$ with a dimension of $1 \times (t_T - 1)$ denotes the training data with one time step shift into the future

$$\mathbf{Y} = [T(t_2) \quad T(t_3) \quad T(t_4) \quad \dots \quad T(t_T)] \quad (4)$$

where t_T denotes the last time step in the training data. $\mathbf{R}$ with a dimension of $(D_r + 3) \times (t_T - 1)$ was composed of the stored values of the input state vector $\mathbf{u}_{in}(t_n)$ and the reservoir nodes $\mathbf{r}(t_n)$ for each time step. $\mathbf{R}$ can be expressed as follows:

$$\mathbf{R} = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ \mathbf{u}_{in}(t_2) & \mathbf{u}_{in}(t_3) & \mathbf{u}_{in}(t_4) & \dots & \mathbf{u}_{in}(t_T) \\ \mathbf{r}(t_2) & \mathbf{r}(t_3) & \mathbf{r}(t_4) & \dots & \mathbf{r}(t_T) \end{bmatrix}. \quad (5)$$

In the prediction phase, the output and input of the RC model were connected, as indicated by the dashed line in Fig. 1. After the initial value $T(t_1)$ was given by the test data, the predicted values $\hat{T}(t_n)$ at $t = t_2, t_3, t_4, \dots$ were calculated sequentially using the following time evolution scheme. As shown in Fig. 1(a), first, I/R received the value for the current time $T(t_n)$ and the pixel electrode voltage for the next time $V_p(t_{n+1})$, which are grouped as the input state vector $\mathbf{u}_{in}(t_n) = \begin{bmatrix} V_p(t_{n+1}) \\ T(t_n) \end{bmatrix}$. Next, the values of the state in the reservoir $\mathbf{r}(t_n)$ were updated as expressed in (1), where $\mathbf{r}(t_n)$ was initialized as $\mathbf{r}(t_1) = \mathbf{0}$. The output $\hat{T}(t_{n+1})$ from R/O was obtained by (2) using the $\mathbf{W}_{out}$ already computed in the training phase. As indicated by the dashed line, the output $\hat{T}(t_{n+1})$ was returned to the input as $T(t_{n+1})$ and formed an input state

vector for the next time step as $\mathbf{u}_{in}(t_{n+1}) = \begin{bmatrix} V_p(t_{n+2}) \\ T(t_{n+1}) \end{bmatrix}$. This process was repeated by the end of the prediction phase.

Fig. 1(b) shows a schematic of the single RC model of liquid crystal cells responding to a design parameter. In this study, we selected the pixel electrode width W (explained in detail in Section II-B) as the input design parameter. The difference from Fig. 1(a) is that the pixel electrode width (difference from the nominal value) is added as an input parameter, which is indicated by ΔW in red in Fig. 1(b). In this study, we set the nominal design value of the pixel electrode width $W = 3 \mu\text{m}$. Correspondingly, the input state vector was changed to

$$\mathbf{u}_{in}(t_n) = \begin{bmatrix} \Delta W \\ V_p(t_{n+1}) \\ T(t_n) \end{bmatrix}.$$

Consequently, the dimensions of the matrices $\mathbf{W}_{in}$, $\mathbf{R}$, and $\mathbf{W}_{out}$ became $D_r \times 4$, $(D_r + 4) \times (t_T - 1)$, and $1 \times (D_r + 4)$, respectively. The reservoir vector $\mathbf{r}(t_n) = [r_1(t_n), r_2(t_n), \dots, r_{D_r}(t_n)]^T$ was initialized as $\mathbf{r}(t_n) = \mathbf{0}$ when the pixel electrode length W was changed in the training data and at the initial time $t = t_1$. The other settings were the same as those of conventional RC models.

Fig. 1(c) shows a schematic of the series-connected multiple RC model with a proven RC model. A well-trained RC model at a specific design value (indicated by blue shading) was connected to the first stage in the series. The value output of the proven RC model in the first stage became the input of the RC model in the second stage. Fig. 1(d) shows a schematic of the parallel-connected multiple RC model with a proven RC model. A well-trained RC model at a specific design value (indicated by blue shading) was connected in parallel. The sum of the outputs from the two reservoirs is the predicted value and the input value of the two reservoirs in the next step. A complete list of the symbols is presented in Table I in Appendix I.

B. Preparation of Training and Test Data

The time-series data for the training and testing the RC model were obtained via numerical simulations using the finite differential method (FDM). The numerical simulation method is not limited to FDM; any generally reliable method, such as the finite element method, can be applied. Experimental data can be used if sufficient data are available. The governing equations solved in the FDM are provided in Appendix II. The fringe field switching (FFS) cell, currently the most popular type of cell in the market, was used as the motif for this study. Fig. 2 shows the structure used to obtain the data. Both the pixel and common electrodes were transparent, with the pixel electrode width denoted as W and the space between pixel electrodes denoted as S under the condition that $S + W = 7.0 \mu\text{m}$. The assumption “ $W + S = \text{const.}$ ” is a condition generally imposed when designing for a determined pixel pitch or when estimating variations in characteristics when the electrode width changes because of manufacturing variations. The other design and material parameters are listed in Appendix III and are identical to those used in our previous study [10]. The incident light came from the common electrode side. The

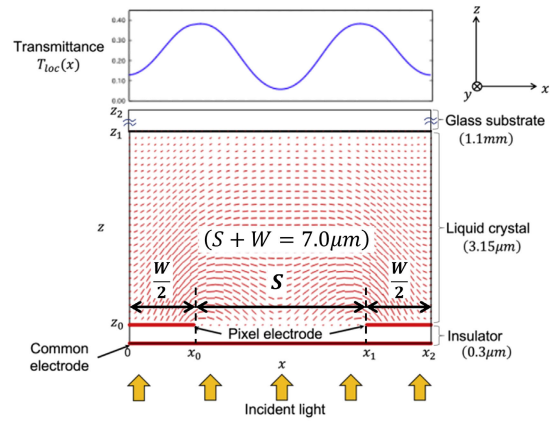


Fig. 2. Structure of a liquid crystal cell with director calculated by FDM simulation to obtain transmittance data for training and testing RC model.

upper panel of Fig. 2 shows the transmittance distribution at each position calculated using the Jones matrix method [17]. The intensity of the incident light was normalized to 1.0. The conditions used in the calculations were periodic boundary conditions on the left and right sides, the Neumann condition above the glass substrate, and the Dirichlet condition on the electrodes.

In this system, the distribution of the liquid crystal director is first calculated, and then the transmittance is determined using the director distribution. Once the distribution of liquid crystal directors is determined at each time, the transmittance T of the liquid crystal cell is obtained by first determining the local transmittance $T_{loc}(x)$ in Fig. 2 and then averaging it over the entire range as follows:

$$T = \frac{1}{x_2 - x_0} \int_{x_0}^{x_2} T_{loc}(x) dx. \quad (6)$$

The voltage of the pixel electrode V_p was initially set to 3.0 V, which was assumed to be the maximum driving voltage. Subsequently, the voltage amplitude of the pixel electrode was changed randomly every 16.7 ms according to a uniform random number distribution from 0.5 to 3.0 V. The voltage polarity was reversed every 16.7 ms. The voltage of the common electrode was always set to 0 V. The total analysis times were 8.35 s (=500 frames) and 1.67 s (=100 frames) for the training and test data, respectively. The input data to the RC model were sampled from the data obtained by the FDM every 500 μs .

The construction of training data depends on the configuration of the RC model, as shown in Fig. 1. For the conventional RC model shown in Fig. 1(a), 8.35 s of data generated using the FDM with the structure $W = 3.0 \mu\text{m}$ ($\Delta W = \pm 0.0 \mu\text{m}$) were used as training data. The training data for RCs with ΔW in input shown in Fig. 1(b)–(d) were assembled using the following procedure.

- 1) For each electrode width ($\Delta W = -2.0, -1.0, \pm 0, +1.0$, and $+2.0 \mu\text{m}$), time-series data of transmittance T were obtained using the FDM. The total analysis time was 8.35 s for each data described above. This data size (=8.35 s) was defined as 100%.
- 2) To evaluate TL ability, data sizes of 40%, 20%, 10%, 2%, ... were generated by obtaining the

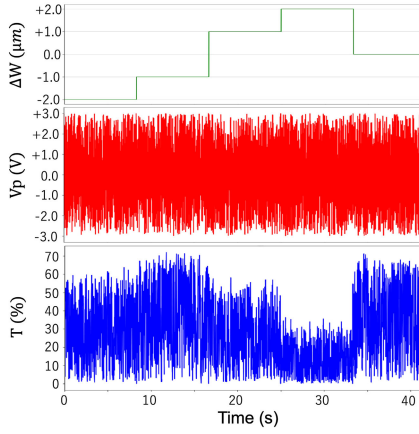


Fig. 3. Time series data generated by FDM for training reservoir (100% data size).

necessary length from the beginning of the 100% dataset.

- 3) The time-series data for each electrode width were concatenated in the time-axis direction. The data image generated in this manner (when each data size was 100%) is illustrated in Fig. 3.

This procedure was performed for convenience in training the RC model with an input port for ΔW . This data image shown in Fig. 3 did not imply that the actual electrode width in LCDs changed with time. As mentioned in Section II-A, because the reservoir $\mathbf{r}(t_n)$ was initialized at the time of the electrode width switch, the data with different electrode widths did not interfere with each other.

Test data used in the prediction phase were prepared with a length of 1.67 s (=100 frames) calculated with nine different electrode widths: test data with trained ΔW ($\Delta W = -2.0, -1.0, \pm 0, +1.0$, and $+2.0 \mu\text{m}$) and test data with untrained ΔW ($\Delta W = -1.5, -0.5, +0.5$, and $+1.5 \mu\text{m}$). The random seeds for generating the voltage amplitude of the pixel electrode were set to values different from those used in the training data.

C. Specification of Proven RC Model

This section presents the specifications of the proven RC model used in the series-connected multiple RC model [Fig. 1(c)] and the parallel-connected multiple RC model [Fig. 1(d)]. The proven RC model was assumed to have been well-trained with a specific design value. In this study, we trained this model in advance using data with a nominal value ($\Delta W = \pm 0.0 \mu\text{m}$), whose length was 8.35 s. Fig. 4 shows the prediction ability of the proven RC model. For this evaluation, as mentioned in Section II-B, we used test data with $\Delta W = \pm 0.0 \mu\text{m}$, whose data length was 1.67 s, generated with a random seed different from that used in training. The root mean square (RMS) error¹ was 0.35%.

¹For the evaluation index of our model, we used a RMS error that is normalized to the average of the reference data as follows:

$$\text{RMS error} = \frac{\sqrt{\frac{1}{m} \sum_{k=1}^m (T_k^{\text{ML}} - T_k^{\text{FDM}})^2}}{\frac{1}{m} \sum_{k=1}^m T_k^{\text{FDM}}} \quad (7)$$

where m represents the total number of data points, and T_k^{ML} and T_k^{FDM} denote the transmittance values from ML and FDM, respectively.

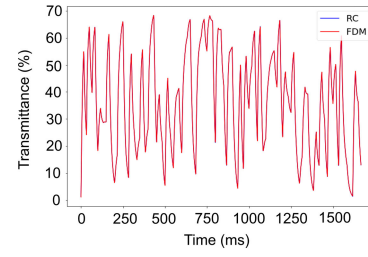


Fig. 4. Prediction ability of proven reservoir, which was well-trained at specific design value ($\Delta W = \pm 0.0 \mu\text{m}$). The hyperparameters : $a = 0.1$, $\rho = 0.8$, $d_i = 0.15$, $D_r = 800$, and $\lambda = 0.001$.

The proven RC model was a very accurate predictor despite being limited to the specific design value.

D. Assumed Use Cases of Our Study

This section provides a hypothetical scenario in which our model can be implemented in the industry.

In the research and development (R&D) phase, prototypes of new devices are developed with representative design values. Subsequently, a macro model is developed for circuit simulation that guarantees accuracy only with the set of design values. At the R&D stage, a conventional RC model without input ports for the design values [Fig. 1(a)] can be sufficient.

In the product design phase, a macro model that responds to design parameters [Fig. 1(b)] is valuable. The purpose of using such a macro model in circuit simulators is to determine the optimal design values that meet the product specifications and predict variations in performance during mass production. Our approach in this study can be used to develop this macro model. The amount of training data required can occasionally be enormous, depending on the range of design values to be changed and their increments. If a sufficient amount of training data does not exist due to time or cost constraints, the proposed TL can be applied by using the model obtained in the R&D phase as a proven model, as shown in Fig. 1(c) and (d).

In the model development flow for circuit simulators described above, capacitance, in addition to transmittance, must be learned via a similar procedure. However, as highlighted in previous studies [4], ensuring prediction accuracy in FFS cell modeling with capacitance models is easier than with transmittance models.

III. RESULTS

A. Single RC Model Responding to Design Parameters

First, we compared the reservoir size D_r to guarantee the prediction accuracy of a conventional RC model [Fig. 1(a)] and a single RC model [Fig. 1(b)] responding to a design parameter. The Ridge coefficient λ was fixed at 1.0, and the other hyperparameters were set to $a = 0.1$, $\rho = 0.8$, and $d_i = 0.15$. The training and prediction data at $\Delta W = \pm 0.0 \mu\text{m}$ were applied to the conventional RC model. Fig. 5 shows the RMS errors at various reservoir sizes D_r . The model was tested using ten different

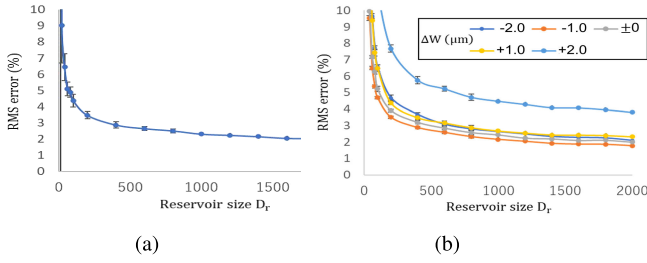


Fig. 5. RMS error during the prediction phase when reservoir size D_r was changed ($\lambda = 1.0$). (a) Conventional RC model trained and tested using the data for $W = 3.0$ μm . (b) Single RC model responding to design parameters.

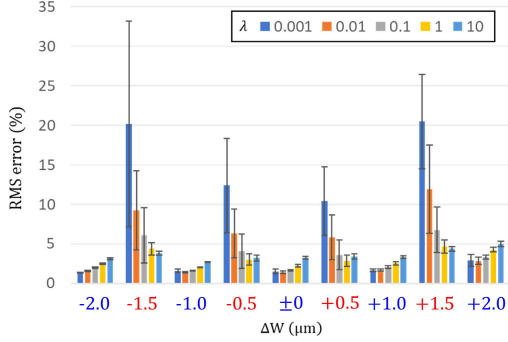


Fig. 6. RMS error when Ridge coefficient λ was changed (D_r was fixed to 1200).

random seeds.² The reservoir size D_r required to achieve the same RMS error was larger in the single RC model responding to a design parameter than in the conventional RC model. For example, the single RC model that covered from $\Delta W = -2.0$ to 2.0 μm with an accuracy within a 5% RMS error required $D_r \geq 800$, whereas the conventional RC model required $D_r \geq 80$.

Next, we examined the influence of the Ridge coefficient λ on the prediction accuracy of the single RC model. In this experiment, the reservoir size D_r was fixed at 1200, and the other hyperparameters were set to $a = 0.1$, $\rho = 0.8$, and $d_i = 0.15$. Fig. 6 plots the RMS error as a function of the Ridge coefficient λ . At the untrained values of $\Delta W = -1.5, -0.5, +0.5$, and $+1.5$ μm (red numbers), the RMS error was reduced by making λ sufficiently large. However, at the trained values of $\Delta W = -2.0, -1.0, \pm 0, +1.0$, and $+2.0$ μm (blue numbers), the RMS error exhibited a slight increase with larger λ .

The impact of the ridge coefficient λ on the actual transmittance waveforms was also investigated. The trained $\Delta W (= +2.0$ $\mu\text{m})$ and untrained $\Delta W (= +1.5$ $\mu\text{m})$ waveforms are presented in Figs. 7 and 8, respectively. The waveforms for the single RC model results in the plots are overlaid for all ten trials with different random seeds. At the trained design values illustrated in Fig. 7, the predicted waveform of the single RC model exhibits a high degree of agreement with the FDM result for both cases of $\lambda = 0.001$ and $\lambda = 10$.

²The elements of W_{res} and W_{in} in RC were determined stochastically; thus, it was not sufficient to evaluate RC performance with a set of specific random numbers. Consequently, ten calculations were performed with different random seeds to assess the robustness of the model. The RMS errors were presented with error bars indicating the “average $\pm$ standard deviation.”

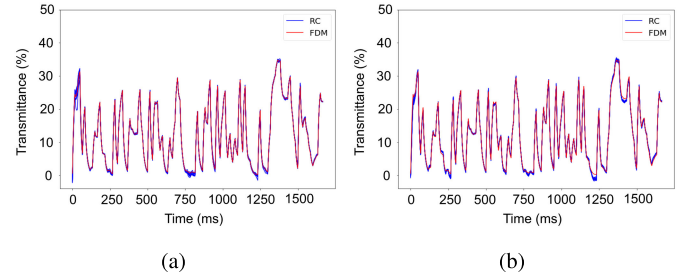


Fig. 7. Transmittance waveform at a trained design value ($\Delta W = +2.0$ μm). (a) $\lambda = 0.001$: RMS error = 2.91 ± 0.75 (%). (b) $\lambda = 10$: RMS error = 5.04 ± 0.34 (%).

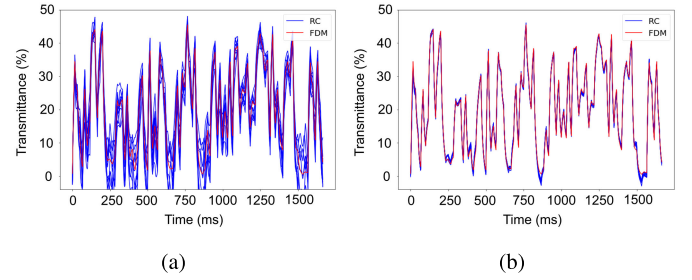


Fig. 8. Transmittance waveform at an untrained design value ($\Delta W = +1.5$ μm). (a) $\lambda = 0.001$: RMS Error = 20.5 ± 5.94 (%). (b) $\lambda = 10$: RMS Error = 4.38 ± 0.31 (%).

Conversely, for the untrained design values depicted in Fig. 8, the larger λ results in a notable improvement in the agreement of the waveforms.

B. TL by Multiple RC Model With a Proven RC

The multiple RC model was evaluated in comparison to the single RC model in terms of its prediction ability. For the multiple RC model, the proven RC model presented in Section II-C and the single RC model evaluated in Section III-A were connected in series or parallel, as illustrated in Fig. 1(c) and (d), respectively. The hyperparameters for the single RC models were set to $D_r = 1200$, $\lambda = 10$, $a = 0.1$, $\rho = 0.8$, and $d_i = 0.15$. Fig. 9 shows the prediction ability at each design value with several training data sizes. The blue lines indicate the outcomes of the single RC models, the orange lines indicate the results of the series-connected multiple RC models, and the green lines indicate the outcomes of the parallel-connected multiple RC models. The dashed red lines indicate the RMS errors of the single RC model trained with a 100% data size. The series-connected multiple RC model exhibited lower RMS errors than the single RC model at all evaluation points. Conversely, the parallel-connected multiple RC model tends to have smaller RMS errors than the single RC model for all electrode widths, except for those with ΔW of $+1.0, +1.5$, and $+2.0$ μm .

Fig. 10 illustrates the required training data sizes for the series- and parallel-connected multiple RC model to achieve an RMS error equivalent to that of the single RC model trained with a 100% data size (Fig. 9) for each design value. The mean value at each evaluation point in Fig. 9 was employed as the data point for the readings, and linear interpolation was used to fill in the gaps between each evaluation point. For example, at $\Delta W = 2.0$ μm , the series-connected multiple RC model demonstrated comparable accuracy with approximately

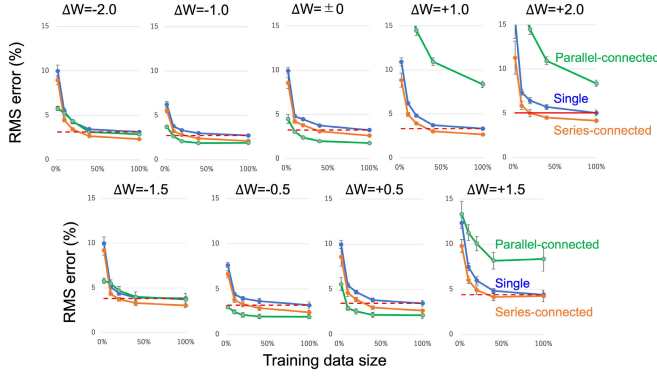


Fig. 9. Prediction ability test results for each design value for various data sizes during training.

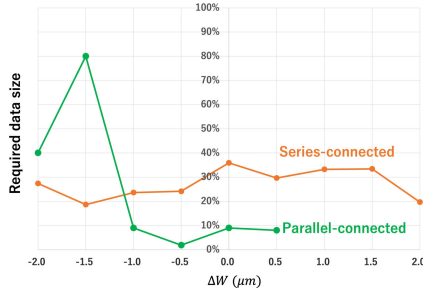


Fig. 10. Data size required for multiple RC model to achieve the same accuracy as when trained on 100% of the data size required for single RC models generated by reading the results represented in Fig. 9.

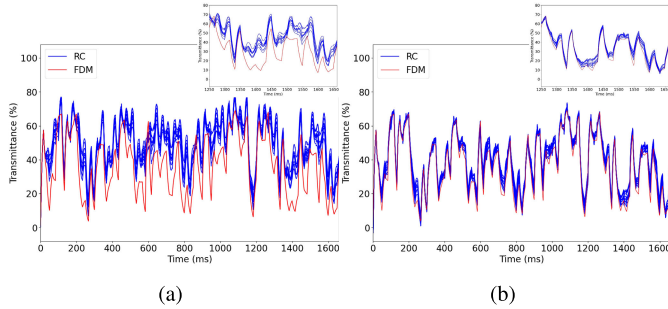


Fig. 11. Transmittance waveform with less training data ($\Delta W = -0.67\%$). (a) Single RC model [RMS error = $45.5 \pm 7.5(\%)$]. (b) Parallel-connected multiple RC model [RMS error = $13.4 \pm 3.8(\%)$].

20% of the training data size of the single RC model. No corresponding point was identified for the conditions of $\Delta W = +1.0, +1.5$, and $2.0 \mu\text{m}$ in the parallel-connected multiple RC model. Fig. 10 illustrates that the series-connected multiple RC model can cover the full range of target electrode widths, from $\Delta W = -2.0 \mu\text{m}$ to $\Delta W = +2.0 \mu\text{m}$, with a training data size of approximately 40% of that required for the single RC model. The parallel-connected multiple RC model demonstrated the capacity to cover the range from $\Delta W = -2.0 \mu\text{m}$ to $\Delta W = +0.5 \mu\text{m}$ with a reduced data size. Furthermore, it exhibited an even smaller data requirement for $\Delta W = -1.0, -0.5, 0$, and $0.5 \mu\text{m}$ than the series-connected multiple RC model. To illustrate this advantage, Fig. 11 compares the outcomes of a prediction test with a single RC model and a parallel-connected multiple RC model with a much smaller data size (0.67%) at $\Delta W = -0.5 \mu\text{m}$. The upper right inset shows an enlarged view after 1250 ms. The parallel-connected

multiple RC model greatly reproduced the FDM results even with the smaller data size.

IV. DISCUSSION

First, we discuss the impact of reservoir size D_r and Ridge coefficient λ on prediction accuracy in an RC model responding to a design value evaluated in Section III-A. As shown in Fig. 5, the reservoir size D_r required to achieve the same RMS error was larger in the single RC model responding to a design parameter than in a conventional RC model. This can be attributed to the fact that the reservoir required greater expressiveness due to the increased number of parameters to learn [18]. To reduce the RMS error at untrained design values, λ must be sufficiently large, as shown in Fig. 6. Nevertheless, an increase in λ slightly increased the RMS error at the trained design values. This could be because the increase in the Ridge coefficient λ resulted in the suppression of overfitting at the trained design values while simultaneously improving the accuracy of the fitted values at the interpolated design values. The optimal Ridge coefficient λ should be determined based on the intended use of the model. This entails considering of whether the objective is to ensure accuracy at specific design values or to guarantee accuracy at interpolated design values. For example, when drafting a design on computer-aided design software, it is preferable to have an accurate model with discrete design values under the minimum grid. In discussing the impact of design value variations on manufacturing, the model needs to guarantee continuous design value variations.

Subsequently, following a review of the limitations inherent in the previous study [12], [13], [14], we discussed the efficacy of the proposed TL approach using multiple RC models. In the RC framework, in (2), the right-hand side comprising the state vector $\mathbf{u}_{in}(t_n)$ and the reservoir vector $\mathbf{r}(t_{n+1})$ and the left-hand side (the predicted value $\hat{T}$) were connected by the weight matrix $\mathbf{W}_{out}$. Therefore, the approach in the previous study was inapplicable if the dimension of the state vector or the reservoir vector differed between the proven RC model and the RC model to be optimized. The dimension of the state vector increased as additional input ports were added to represent the response to new parameters (the electrode width in this study). The dimension of the reservoir vector increased when additional complex inferences were required while maintaining accuracy. This was not only the case when input ports were added but also when the range of parameters covered was expanded. As shown in Fig. 5(a), the required dimension of the reservoir vector was significantly increased by covering the electrode width $\Delta W = +2.0 \mu\text{m}$. Therefore, we developed a more flexible TL approach using multiple RC models.

As shown in Fig. 10, except for some conditions of the parallel-connected multiple RC model ($\Delta W = +1.0, +1.5$, and $+2.0 \mu\text{m}$), the training data size required to achieve the same accuracy was significantly less for the multiple RC model than for the single RC model. This suggests that the “knowledge” of a highly accurate proven RC model with a specific design value ($\Delta W = \pm 0.0 \mu\text{m}$) can be successfully transferred to other design values. In general, the success factor of TL is efficient generation of a learning model using the similarities between different tasks. Fig. 12 plots the results of

TABLE II
PARAMETERS USED IN THE FDM SIMULATION

$\varepsilon_{\perp}$	4.1	$\varepsilon_{\parallel}$	18.9
n_o	1.490	n_e	1.615
K_{11}	8.5 pN	K_{22}	5.7 pN
K_{33}	12.3 pN	γ_1	0.117 Pa s
ε_{ins}	6.5	ε_{glass}	5.0
$\phi(z = z_0, z_1)$	5°	$\theta(z = z_0, z_1)$	1°
λ_0	550 nm		

a rule of Einstein's notation. In solving (A.1) and (A.2), $|\mathbf{n}(x, z)| = 1$ is assumed because only the orientation of $\mathbf{n}(x, z)$ is meaningful. In (A.1), ε_0 denotes the dielectric constant of the vacuum, and E_j denotes the electric field, and ε_{ij} denotes the dielectric tensor of the liquid crystal, which is expressed as follows:

$$\varepsilon_{ij} = \varepsilon_{\perp} \delta_{ij} + (\varepsilon_{\parallel} - \varepsilon_{\perp}) n_i n_j \quad (\text{A.3})$$

where $\varepsilon_{\parallel}$ and $\varepsilon_{\perp}$ denote the dielectric constants of liquid crystals along parallel and perpendicular to the liquid crystal director, respectively; n_i denotes the component of the liquid crystal director; δ_{ij} represents the Kronecker delta function. In (A.2), γ_1 denotes the rotation viscosity constant, and M_i is generally called a molecular field in analogy to magnetism and is expressed as follows:

$$M_i = \left(\frac{\partial F}{\partial n_{i,j}} \right)_{,j} - \frac{\partial F}{\partial n_i} \quad (\text{A.4})$$

where the subscript j preceded by a comma denotes differentiation for the j th coordinate, and F represents the total Gibbs free energy consisting of the following three terms:

$$F = F_{\text{elastic}} + F_{\text{dielectric}} \quad (\text{A.5})$$

$$F_{\text{elastic}} = \frac{1}{2} K_{11} (\nabla \cdot \mathbf{n})^2 + \frac{1}{2} K_{22} (\mathbf{n} \cdot (\nabla \times \mathbf{n}) + q_0)^2 + \frac{1}{2} K_{33} (\mathbf{n} \times (\nabla \times \mathbf{n}))^2 \quad (\text{A.6})$$

$$F_{\text{dielectric}} = -\frac{1}{2} (\mathbf{E} \cdot \mathbf{D}) \quad (\text{A.7})$$

where K_{11} , K_{22} , and K_{33} denote the Frank elastic constants for splay, twist, and bend, respectively; q_0 denotes a constant determined by the material described by $q_0 = (2\pi/P)$, where P represents the chiral pitch of the liquid crystal. This study assumed $q_0 = 0$ for the nonchirality material used in general FFS. The components of the displacement field vector $\mathbf{D}$ can be expressed as $D_i = \varepsilon_0 \varepsilon_{ij} E_j$.

APPENDIX III PARAMETERS IN FDM SIMULATION

Table II lists all material parameters used in the FDM simulation. In Table II, $\varepsilon_{\perp}$ and $\varepsilon_{\parallel}$ denote the dielectric constants of a liquid crystal perpendicular and parallel to the liquid crystal director, respectively; n_o and n_e denote the refractive indices of polarized light perpendicular to the liquid crystal director and parallel to the director, respectively; K_{11} , K_{22} , and K_{33} denote the Frank elastic constants for splay, twist, and bend, respectively; γ_1 denotes the rotation viscosity constant;

ε_{ins} and ε_{glass} denote the dielectric constants of the insulator and the glass substrate in Fig. 2, respectively; ϕ and θ denote the twist and tilt angles, respectively; ϕ denotes the angle with the y-axis, and θ denotes the angle with the x-y plane. For the optical simulation, the transmission axis of the polarizer is aligned with the direction of the liquid crystal director at $z = z_0$, and that of the analyzer is aligned perpendicular to the direction of the liquid crystal director at $z = z_1$. λ_0 denotes the wavelength of the incident light.

REFERENCES

- [1] K. Yin et al., "Advanced liquid crystal devices for augmented reality and virtual reality displays: Principles and applications," *Light, Sci. Appl.*, vol. 11, no. 1, p. 161, May 2022.
- [2] Y.-H. Wu et al., "Breaking the limits of virtual reality display resolution: The advancements of a 2117-pixels per inch 4K virtual reality liquid crystal display," *J. Opt. Microsyst.*, vol. 3, no. 4, Oct. 2023, Art. no. 041208.
- [3] C. Kim, A. Klement, E. Park, J. Han, L. Rao, and J. Zhuang, "6-2: Invited paper: High-ppi fast-switch display development for oculus quest 2 VR headsets," in *SID Symp. Dig. Tech. Papers*, 2022, vol. 53, no. 1, pp. 40–43.
- [4] M. Watanabe, K. Ishihara, T. Tsuruma, Y. Iguchi, Y. Nakajima, and Y. Maki, "Macro-modeling of liquid crystal cell with VerilogA," in *Proc. IEEE Int. Behav. Modeling Simulation Workshop*, Sep. 2007, pp. 132–137.
- [5] D.-H. Jeon, J.-M. Kim, S.-H. Lee, and S.-W. Lee, "A simple overdrive technology using behavioral circuit model for liquid crystal displays," *IEEE Trans. Electron Devices*, vol. 63, no. 6, pp. 2436–2441, Jun. 2016.
- [6] Y. Cho, C. Park, J.-M. Kim, S.-H. Lee, and S.-W. Lee, "A behavioral circuit model of active-matrix liquid crystal displays for optical response simulation," *IEEE Trans. Electron Devices*, vol. 59, no. 5, pp. 1430–1438, May 2012.
- [7] B. Schrauwen, D. Verstraeten, and J. M. V. Campenhout, "An overview of reservoir computing: Theory, applications and implementations," in *Proc. 15th Eur. Symp. Artif. Neural Netw.*, 2007, pp. 471–482.
- [8] J. Pathak, Z. Lu, B. R. Hunt, M. Girvan, and E. Ott, "Using machine learning to replicate chaotic attractors and calculate Lyapunov exponents from data," *Chaos, Interdiscipl. J. Nonlinear Sci.*, vol. 27, no. 12, Dec. 2017, Art. no. 121102.
- [9] K. Nakai and Y. Saiki, "Machine-learning inference of fluid variables from data using reservoir computing," *Phys. Rev. E, Stat. Phys. Plasmas Fluids Relat. Interdiscip. Top.*, vol. 98, no. 2, Aug. 2018, Art. no. 023111.
- [10] M. Watanabe, K. Kotani, and Y. Jimbo, "Macro modeling of liquid crystal cell using machine learning method: Reservoir computing approach," *Jpn. J. Appl. Phys.*, vol. 62, no. 2, Feb. 2023, Art. no. 027001.
- [11] R. B. Meyer, "Piezoelectric effects in liquid crystals," *Phys. Rev. Lett.*, vol. 22, no. 18, pp. 918–921, May 1969.
- [12] M. Inubushi and S. Goto, "Transfer learning for nonlinear dynamics and its application to fluid turbulence," *Phys. Rev. E, Stat. Phys. Plasmas Fluids Relat. Interdiscip. Top.*, vol. 102, no. 4, Oct. 2020, Art. no. 043301.
- [13] X. Chen, B. T. Nadiga, and I. Timofeyev, "Predicting shallow water dynamics using echo-state networks with transfer learning," *GEM-Int. J. Geomath.*, vol. 13, no. 1, p. 20, Dec. 2022.
- [14] I. Bauwens, K. Harkhoe, P. Bienstman, G. Verschaffelt, and G. Van der Sande, "Transfer learning for photonic delay-based reservoir computing to compensate parameter drift," *Nanophotonics*, vol. 12, no. 5, pp. 949–961, Mar. 2022.
- [15] H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks-with an erratum note," *German Nat. Res. Center Inf. Technol., Bonn, Germany, Tech. Rep.* 148, 2001, vol. 148, no. 34, p. 13.
- [16] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, Feb. 1970.
- [17] P. Yeh and C. Gu, *Optics of Liquid Crystal Displays*, vol. 67. Hoboken, NJ, USA: Wiley, 2009.
- [18] M. Lukoševičius, "A practical guide to applying echo state networks," in *Neural Networks: Tricks of the Trade*. Cham, Switzerland: Springer, 2012, pp. 659–686.